OFFICIAL SOURCE CODE DOCUMENTATION

# Faculty Feedback & Appraisal System

Complete Core Architecture, Scoring Engine, API & UI Codebase

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|---|---|
| **GitHub Repo:** | `https://github.com/ayushgautam50605-ship-it/faculty-feedback-system` |
| **Branch:** | `main` |
| **Target Stack:** | `Node.js, Express, SQLite, React (Vite), Tailwind CSS` |
| **Department:** | `Information Technology (Semester 5)` |
| **Generated Date:** | `November 2, 2026` |

## PROJECT ARCHITECTURAL SUMMARY

The Faculty Feedback & Appraisal System is an end-to-end academic analytics platform that streamlines semester-end faculty evaluations. It replaces manual paper forms with an automated, transparent appraisal scoring pipeline. The system enforces strict role-based access control (HOD, Faculty, Student), prevents duplicate submissions, protects student anonymity, and computes weighted multi-criteria appraisal grades (Satisfactory, Needs Improvement, Excellent) with real-time analytics.

Express REST API | SQLite + better-sqlite3 | JWT + Bcrypt Auth | React 18 + Vite | Tailwind CSS | Lucide Icons | Recharts Analytics

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BACKEND / DATABASE

# 1. Database Schema & Configuration

📁 `backend/config/db.js`

SQLite database setup via better-sqlite3 with foreign key enforcement and 8 normalized tables (users, students, faculty, subjects, criteria, feedback_periods, feedback, feedback_ratings, appraisal_config, appraisal).

```
1  // File not found at: backend/config/db.js
```

BACKEND / CORE LOGIC

## 2. Appraisal Scoring Engine

📁 `backend/utils/appraisal.js`

The mathematical appraisal engine that computes criterion averages, weighted ratings, normalized 0–100 appraisal percentages, performance grades, and upserts faculty appraisals.

```
1  // File not found at: backend/utils/appraisal.js
```

BACKEND / SECURITY

## 3. JWT Authentication & Role Authorization Middleware

📁 `backend/middleware/auth.js`

Bearer token verification and role-based access control (RBAC) ensuring students, faculty, and HOD can only access their authorized endpoints.

```
1  // File not found at: backend/middleware/auth.js
```

BACKEND / CONTROLLERS

## 4. Auth Controller & Student Self-Registration

📁 `backend/controllers/authController.js`

Handles user login, token signing, and self-registration for new students enrolling into the system with hashed passwords.

```
1  // File not found at: backend/controllers/authController.js
```

BACKEND / CONTROLLERS

# 5. Feedback Controller

📁 `backend/controllers/feedbackController.js`

Validates criterion ratings (1-5), prevents duplicate feedback per period, manages atomic transaction commits for multi-criterion ratings, and tracks student progress.

```
1  // File not found at: backend/controllers/feedbackController.js
```

BACKEND / CONTROLLERS

## 6. Analytics & Performance Aggregation Controller

📁 `backend/controllers/analyticsController.js`

Aggregates student feedback into real-time KPIs, department-wide rating averages, faculty rankings, and radar/bar chart datasets for the HOD dashboard.

```
1  // File not found at: backend/controllers/analyticsController.js
```

BACKEND / ENTRYPOINT

## 7. Express API Server Entrypoint

📁 `backend/server.js`

Main Express application configuration, CORS handling, JSON body parsing, route registration, and centralized error handling.

```
1  // File not found at: backend/server.js
```

BACKEND / SCRIPTS

## 8. Faculty & Subject Population Script

📁 `backend/add_faculty.js`

Automated provisioning script that registers Information Technology faculty members and links them to their respective semester 5 subjects.

```
1  // File not found at: backend/add_faculty.js
```

BACKEND / SCRIPTS

## 9. Student Population Script

📁 `backend/add_students.js`

Registers initial student cohort in the Information Technology department with secure bcrypt credentials.

```
1  // File not found at: backend/add_students.js
```

FRONTEND / STATE

## 10. Global Authentication Context

📁 `frontend/src/context/AuthContext.jsx`

React Context managing global user state, token persistence in localStorage, student registration dispatch, and login/logout lifecycles.

```
1  // File not found at: frontend/src/context/AuthContext.jsx
```

FRONTEND / PAGES

## 11. Student Feedback Submission Component

📁 `frontend/src/pages/student/FeedbackForm.jsx`

Interactive student feedback interface featuring 1–5 rating inputs for every active criterion, optional qualitative comments, and submission confirmation.

```
1  // File not found at: frontend/src/pages/student/FeedbackForm.jsx
```

FRONTEND / PAGES

## 12. Student Dashboard

📁 `frontend/src/pages/student/StudentDashboard.jsx`

Student overview showing pending and completed subject reviews, completion progress bar, and academic semester status.

```
1  // File not found at: frontend/src/pages/student/StudentDashboard.jsx
```

FRONTEND / PAGES

## 13. Faculty Performance Dashboard

📁 `frontend/src/pages/faculty/FacultyDashboard.jsx`

Faculty dashboard displaying calculated appraisal score, overall rating, criteria-by-criteria breakdown charts, and anonymous student comments.

```
1  // File not found at: frontend/src/pages/faculty/FacultyDashboard.jsx
```

FRONTEND / PAGES

## 14. HOD Analytics & Appraisal Dashboard

📁 `frontend/src/pages/hod/HodDashboard.jsx`

Comprehensive administration dashboard featuring response completion rates, department rankings, semester comparison trends, and faculty scores.

```
1  // File not found at: frontend/src/pages/hod/HodDashboard.jsx
```